CSE 499B: Project Update 05 | Section: 01 | Group: 07 | Members: Ashraful Islam (2212669042), Tanushree Das (2212225042).

Project Progress Report: Arsenic Skin Detection using Machine Learning

I. Overall Summary of Progress

This week, we successfully completed our project. We finalized both the frontend and backend parts of the system and made everything work together properly. The system can now take a skin image as input and predict whether the person is affected by arsenic or not.

We also implemented the chatbot, which helps users by giving suggestions and basic guidance based on the result. The UI (Dashboard, Login, Registration, Logout) is fully connected with the backend, and the whole system is now working smoothly.

Overall, this week was focused on final integration, testing, and completing the project.

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II. Individual Contributions this Week

A. Ashraful Islam (System Integration & Final Implementation)

- Completed most of the final work.
- Integrated frontend (HTML, CSS) with backend (Flask/Python).
- Implemented the image prediction system.
- Connected the chatbot with the main system.
- Tested the full system workflow (Login → Dashboard → Prediction → Chatbot).
- Fixed bugs and improved performance.

B. Tanushree Das (Testing & Final Improvements)

- Helped in testing the system.
- Checked UI design and fixed small issues.
- Assisted in chatbot improvement.
- Helped in final review and project completion.

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IV. Challenges We Faced

During the final stage, we faced some problems in connecting the frontend with the backend. Sometimes the image upload and prediction did not work correctly, which required debugging. Integrating the chatbot with the system was also a bit difficult, as we needed to make sure it gives proper responses.

We solved these issues by testing step by step and fixing errors. It took time, but finally everything worked correctly.

V. Conclusion

In conclusion, our project is a simple and useful system that can detect arsenic from a skin image. The user can easily register, login, upload an image, and get a result showing whether they are affected or not. The system is designed to be user-friendly so that anyone can use it without difficulty.

It also includes a chatbot that gives suggestions and basic guidance based on the result, which makes the system more helpful. Overall, our project combines UI design, backend, and machine learning to create a complete and working solution.